



11104 - Galactic Entourage: A Spectroscopic Study of Satellite Galaxies at Cosmic Noon

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	satellite_candidates	NIRSpec MultiObject Spectroscopy	(1) primary_catalog_z3sigma

ABSTRACT

Satellite galaxies offer powerful tests of galaxy evolution and environmental quenching models, yet at high redshift they remain virtually unexplored. At cosmic noon ($1 < z < 3$), when galaxies and halos were rapidly assembling, models predict much lower quenched fractions than at $z \sim 0$, as well as evolution in their mass and radial distributions; nevertheless, these predictions remain largely observationally untested. Low-mass satellites ($< 1/10$ the mass of their hosts) at $z > 1$ are now easily resolved in deep-field JWST imaging, opening a new regime for studying their demographics and star-formation activity. However, photometric redshift uncertainties can lead to substantial contamination and systematic biases, and no systematic spectroscopic study of these faint satellites has ever been conducted. We propose a pilot, single-pointing NIRSpec MSA program to obtain $R \sim 100$ PRISM/CLEAR spectra for all most-likely satellite candidates with F200W magnitudes < 27.7 AB mag around five massive ($M_{\text{star}} > 10^{11} M_{\text{sun}}$) host galaxies at $1 < z < 3$. These data will provide the first spectroscopic sample of low-mass satellites at cosmic noon, enabling robust membership

confirmation, mapping of radial distributions, and measurement of star-formation rates. This will allow us to begin calibrating high- z photometric studies of satellites, measuring the structural properties of satellite systems, directly testing models of environmental quenching, and establishing a foundation for statistical spectroscopic analyses of satellites in the early universe.

OBSERVING DESCRIPTION

Instrument: We propose to use a single pointing of the NIRSpec MSA to observe the satellite galaxies of five massive (stellar masses $>10^{11}$ Msun) galaxies at $1 < z < 3$. These systems were selected from public PRIMER and MINERVA imaging in the UDS field. We choose the PRISM/CLEAR disperser–filter combination for its efficiency and broad spectral coverage, enabling detection of H α and the Balmer break across the full $1 < z < 3$ range, the latter being particularly important for securing the redshifts of quenched galaxies. This broad wavelength coverage also ensures robust redshifts for identification of higher-redshift interlopers. The $R \sim 100$ PRISM spectra are fully sufficient for resolving the ~ 2000 km/s necessary to confirm satellite association with their hosts.

Depth: Our maximum required exposure time is set to achieve continuum S/N ~ 5 , ensuring robust redshift measurements for quiescent satellites down to our magnitude limit (F200W magnitude of 27.7 AB mag, corresponding to $\sim 10^8$ Msun). We run the ETC on a public PRISM spectrum of a stellar mass $\sim 10^{8.5}$ Msun quiescent galaxy at $z \sim 1.5$ (PID 6368). Renormalizing the spectrum to different F200W magnitudes, we find that the continuum can be detected at 5 σ for F200W magnitudes of 27.1, 27.5, and 27.7 AB mag with exposure times of 8.0 ks, 16.0 ks, and 24.0 ks, respectively. For a star-forming galaxy, assuming no detectable continuum, using the ETC we find that 5 σ detections of H α can be achieved for line fluxes of 3.2, 2.2 and 1.8×10^{-18} erg/cm 2 /s for the same three exposure times, roughly 0.2 Msun/yr or the star-forming main sequence for 10^8 Msun galaxies at $z \sim 1.5$ (Popesso et al. 2023, Merida et al. 2025).

Observational strategy: The satellite systems are too dense to simultaneously fit all targets on a single MSA configuration, but are much too extended for an IFU pointing. MPT modeling suggests that four configurations is sufficient to confirm redshifts for all highest-priority satellite candidates and $\sim 90\%$ of medium-priority candidates (as described below). We use a three-tier observing strategy to allow us to reach the same $\sim 10^8$ Msun limiting mass for faint quiescent satellites (continuum) as for faint star-forming satellites (lines). Satellite candidates with either F200W magnitudes < 27.1 mag or estimated H α fluxes $> 3.2 \times 10^{-18}$ erg/cm 2 /s will be observed in a single MSA configuration. Those with F200W magnitudes > 27.5 mag or H α fluxes $< 1.8 \times 10^{-18}$ erg/cm 2 /s will be observed in three configurations, and sources in between will be observed in two configurations. Each MSA configuration will be observed for an exposure time of 8009s with a 3-shutter nodding pattern. This guarantees 24 ks depth for the faintest sources while minimizing excess exposure on bright targets and retaining the flexibility to cover all targets.

JWST Proposal 11104 (Created: Friday, March 13, 2026, 6:01:17PM Eastern Standard Time) - Overview

Target priority: Satellite candidates are defined as sources within a projected separation of 300 kpc ($\sim R_{\text{vir}}$ at $z \sim 1$) from the selected hosts, whose photometric redshifts are consistent with those of their hosts to within 3σ . Candidates above our magnitude limit and within projected separations of 150 kpc ($\sim 0.5 R_{\text{vir}}$ at $z \sim 1$) whose redshifts agree within 1σ represent our highest-priority targets, for which we expect 100% completeness under our observing strategy. Candidates consistent within $2\text{--}3\sigma$ are assigned medium priority, with an expected $\sim 90\%$ completeness from MPT simulations. Those located between 150 kpc and 300 kpc ($\sim R_{\text{vir}}$ at $z \sim 1$) constitute our low-priority/filler targets.

For proposal optimization purposes, we have used a catalog of our high- and medium- priority targets only, to ensure the feasibility of our observing strategy. As such, the proposed MSA configurations are not final, and the total number of targets per MSA configuration will increase when our low-priority targets are included in the MPT input catalogs.

Proposal 11104 - Targets - Galactic Entourage: A Spectroscopic Study of Satellite Galaxies at Cosmic Noon

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	primary_catalog_z3sigma	RA: 02 17 19.8625 (34.3327604d) Dec: -05 11 58.69 (-5.19964d) Equinox: J2000		
	Comments: Description=[]				

Proposal 11104 - Observation 1 - Galactic Entourage: A Spectroscopic Study of Satellite Galaxies at Cosmic Noon

Observation	Proposal 11104, Observation 1: satellite_candidates										Fri Mar 13 23:01:17 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRSpec MultiObject Spectroscopy										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	primary_catalog_z3sigma	RA: 02 17 19.8625 (34.3327604d) Dec: -05 11 58.69 (-5.19964d) Equinox: J2000								
Acquisition	Comments: Description=[]										
	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
Template	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
Reference Stars	MSATA	false	No	MSA Center	primary_catalog_z3sigma (142 sources)	primary_catalog_z3sigma (142 sources)	jwst-nirspec-prism	1.5			

Proposal 11104 - Observation 1 - Galactic Entourage: A Spectroscopic Study of Satellite Galaxies at Cosmic Noon

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	34.326242041666 67 Degrees - 5.191368055555 29 Degrees	195.00056857575 873			3	9	8009.301
	2	1 (PRISM/CLEAR)	c2	3 Shutter Slitlet	34.32613375 Degrees - 5.191770555555 36 Degrees	195.00057900366 82			3	9	8009.301
	3	1 (PRISM/CLEAR)	c3	3 Shutter Slitlet	34.323225583333 33 Degrees - 5.189700277777 74 Degrees	195.00083546351 64			3	9	8009.301
	4	1 (PRISM/CLEAR)	c4	3 Shutter Slitlet	34.320259416666 666 Degrees - 5.191065555555 54 Degrees	195.00110354204 654			3	9	8009.301
Special Requirements	MSA Planned Aperture PA 195.0000 to 195.0000 Degrees (V3 56.4254303 to 56.4254303)										